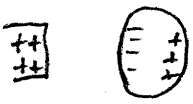
HW 6

1 a). Perhaps nylon losing $e^-$, rubber gaining

b) 

Conductor $e^-$ gather near insulator

c) The ground $e^-$ are attracted to the positive insulator. Sphere is net negative now

d) The $e^-$ would go back to ground, and the sphere would be neutral again

2 a) Earth crust, mantle $\left\{ \overset{16}{O}, \overset{24}{Mg}, \overset{28}{Si}, \overset{55}{Fe}, \overset{27}{Al} \right\}$ say avg 23 g/mol $\}$ 67%

core mostly Fe (85%), so say 55 g/mol } 33%

total ~34 g/mol so $\frac{6\times10^{24}}{.034} = 1.8\times10^{26}$ moles;

~16 protons per mole $\Rightarrow$ $1.7\times10^{31}$ protons

Sun 73.4% H, 25% He, rest heavier, avg 1.7 g/mol

so $\frac{2\times10^{30}}{.0017} = 1.17\times10^{33}$ moles, 1.25 protons/mole,

$\Rightarrow$ $8.8\times10^{36}$ protons

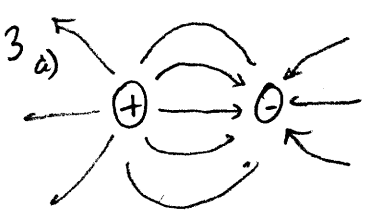
b) Coulomb force $\frac{9\times10^9 (1.7\times10^{31})(1.6\times10^{-19})(8.8\times10^{36})(1.6\times10^{-19})}{(1.5\times10^{11})^2}$

$= 1.5\times10^{36}$ N

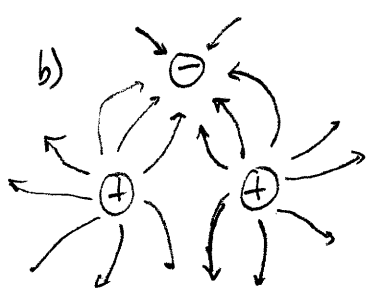
Grav. force $= \frac{(6.67\times10^{-11})(2\times10^{30})(6\times10^{24})}{(1.5\times10^{11})^2} = 3.55\times10^{22}$ N

c) $\frac{50\text{ C}}{1.6\times10^{-19}\text{ C/e}} = 3.125\times10^{20}\ e^- \Rightarrow \frac{3.125\times10^{20}}{6.02\times10^{23}\ e^-/\text{mole}} = 5.19\times10^{-4}$ moles

3 a)



b)



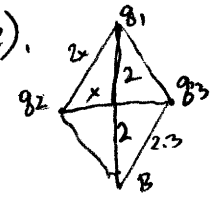
c). Call the charges 1 (negative), 2 (left +), 3 (right +).

$\vec{E}_2$ and $\vec{E}_3$ cancel at A by symmetry.

$\vec{E}_1 = \frac{kq_1}{r_{1A}^2} = \frac{9\times10^9 \cdot 30\times10^{-9}}{(.02)^2} = 6.75\times10^5$ N/C up

3d) $\vec{F} = q\vec{E} = 0.0135$ N down

e).



$x^2+4=(2x)^2$

$x = 1.15$ cm

$\vec{E}_1 = \frac{(9\times10^9)(30\times10^{-9})}{(.04)^2} = 1.69\times10^5$ up or $\hat{y}$

$\vec{E}_2 = \frac{(9\times10^9)(30\times10^{-9})}{(.0239)^2}\left(\hat{x}\cdot\frac{1.15}{2.3} - \hat{y}\cdot\frac{2}{2.3}\right)$

$\vec{E}_3 = \frac{(9\times10^9)(30\times10^{-9})}{(.0239)^2}\left(-\hat{x}\cdot\frac{1.15}{2.3} - y\frac{2}{2.3}\right)$

$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = 0\hat{x} - 7.08\times10^5\ \hat{y}$

4a). $\vec{E} = \vec{E}_1 + \vec{E}_2 = \left(\frac{kQ}{(r+d/2)^2} - \frac{kQ}{(r-d/2)^2}\right)\hat{x}$

b). atomic; $Q = 1.6\times10^{-19}$, $d = 0.5$ nm

$E(r=5d) = 2.35\times10^7$ N/C

$E(r=100d) = 2880$ N/C

falls off fast

only +Q:

$E(r=5d) = 4.76\times10^9$

$E(r=100d) = 144000$

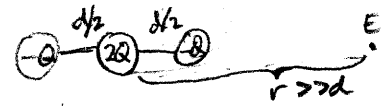
decreases slower

c). $\frac{kQ}{(r+d/2)^2} = \frac{kQ}{r^2(1+\frac{1}{2}\frac{d}{r})^2} \approx \frac{kQ}{r^2}\left(1 - \frac{d}{r} + O\left(\frac{d}{r}\right)^2\right)$

$-\frac{kQ}{(r-d/2)^2} = -\frac{kQ}{r^2}\left(1 - \frac{d}{r} + O\left(\frac{d}{r}\right)^2\right)$

$\therefore \vec{E} \approx -\frac{2kQd}{r^3}$ $\left(E \sim \frac{1}{r^3}\right)$ unlike point $E \sim \frac{1}{r^2}$.

d).



$E_{left} = \frac{-kQ}{(r+\frac{d}{2})^2} \approx -\frac{kQ}{r^2}\left(1 - \frac{d}{r} + \frac{3}{4}\left(\frac{d}{r}\right)^2 + \dots\right)$

$E_{middle} = \frac{+k(2Q)}{r^2}$

all in $+\hat{x}$ dir

$E_{right} = \frac{-kQ}{(r-\frac{d}{2})^2} = \frac{-kQ}{r^2}\left(1 + \frac{d}{r} + \frac{3}{4}\left(\frac{d}{r}\right)^2 + \dots\right)$

$E_{tot} = -\frac{3kQd^2}{2r^4}$ $\left(E \sim \frac{1}{r^4}\right)$

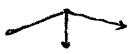
5b) Using a circle of radius 1 km,

$A = \pi(1000)^2$

$$E = \frac{100}{8.85\times10^{-12}\,\pi(1000)^2} = 3.6\times10^6$$

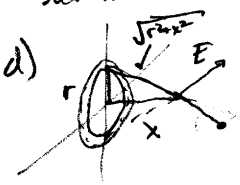
c) Yes, $E > 3\times10^6$

a). Close: For:



The components add better far away, but are further. Effects cancel.

d)




Vertical components from ring cancel by symmetry.

Horizontal component is $\frac{k\,dq_{ring}}{r^2+x^2}\cdot\frac{x}{\sqrt{r^2+x^2}}$.

$dq$ for the ring is $\sigma\cdot 2\pi r\,dr$.

Adding up all the rings:

$$E = \int_0^\infty \frac{k\cdot 2\pi r\,dr\,x}{(r^2+x^2)^{3/2}}\,dr = \frac{2\pi x\sigma}{4\pi\varepsilon_0}\left[(r^2+x^2)^{-1/2}\right]_0^\infty$$

$$= \frac{\sigma x}{2\varepsilon_0}\cdot\frac{1}{x} = \frac{\sigma}{2\varepsilon_0} = \frac{Q}{2\varepsilon_0 A}$$

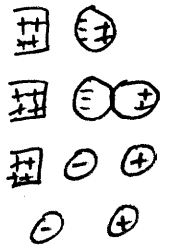
## Quiz 4

1a. Step 1.

Step 2

Step 3

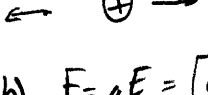
Step 4



So negative on A, pos on B.

b. electrons

2 a)



b). $F = qE = \boxed{6\times10^{-4}\text{ N right}}$

c) No, $20000 < 3\times10^6$

no ionization of air

## HW 7

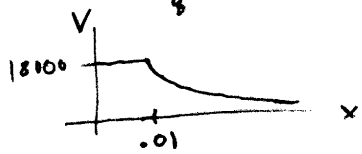
1a. 9V, more charge for larger potential difference.

b. $V = \frac{U}{q} = \frac{kQq}{rq} = \frac{kQ}{r}$. Definition of potential

c. $V = \frac{9\times10^9\cdot 20\times10^{-9}}{.01} = 18000\text{ V}$

d. Effects cancel, V same. More charge but spread further apart.

e. $\Delta V = \frac{\Delta U}{q}$, since $\Delta U = 0$, $\Delta V = 0$.




2a. $W = F\cdot d$. Here, $F = qE$ so $W = qEd = q(1.6\times10^8)$ J

$\Delta V = \frac{\Delta U}{q} = \boxed{1.6\times10^8\text{ V}}$

b. $\frac{170}{3\times10^6} = 5.67\times10^{-5}\text{ m} = .0567\text{ mm}$

c. The charge can move, so E does not remain constant (recall $E = \frac{Q}{\varepsilon_0 A}$) even as $\Delta V$ does, in this case

3a. $\frac{1.5\text{ C}}{1.6\times10^{-19}\text{ C/e}^-} = 9.4\times10^{18}\ e^-$

b. $r = .814\text{ mm}$, $A = \pi r^2 = 2.08\times10^{-6}\text{ m}^2$

Bonus: $\frac{3}{2}kT = \frac{1}{2}m\langle v^2\rangle$ so rms $v = \sqrt{\frac{3kT}{m}} = 1.15\times10^5$ m/s $= 115$ km/sec.

c. $8.5\times10^{28}\ e^-/\text{m}^3$

Bonus: Cu 8.96 g/cc $= 8.96\times10^6$ g/m³

copper 63.5 g/mole so 1m³ has $\frac{8.96\times10^6}{63.5} = 1.41\times10^5$ moles of Cu atoms

each atom contributes 1 conduction electron. Thus

$n = 1.41\times10^5\text{ moles/m}^3\cdot 6.02\times10^{23}\text{ e}^-/\text{mole} = 8.49\times10^{28}\ e^-/\text{m}^3$

d. Need $9.4\times10^{18}\ e^-$ per second. Have $8.5\times10^{28}\cdot Av$ per second.

so $v = \frac{9.4\times10^{18}}{8.5\times10^{28}\cdot A} = 5.32\times10^{-5}$ m/s $= \boxed{.053\text{ mm/sec. Small}}$

Bonus. $x = vt = 5.32\times10^{-5}\cdot\frac{1}{120} = 4.43\times10^{-7}$ m. This is around 400 interatomic distances

4 a. More charge built up; larger E possible

b. More collisions in skinny, long wire

c. $R = 20\,\Omega$, consistent for these V's

d. $R = 5\,\Omega$, less collisions when cold

e. $P = IV = 720$ W. To heat + light at first, $P = 2880$ W. To heat element

5. Spherical $\vec{\nabla}f = \frac{\partial f}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial f}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial\phi}\hat{\phi} \Rightarrow -\vec{\nabla}\left(\frac{kq}{r}\right) = \frac{kq}{r^2}\hat{r} + 0\hat{\theta} + 0\hat{\phi} = \frac{kq}{r^2}\hat{r} = \vec{E}$ QED

## Quiz 5

1a. $I_1 = \frac{V}{R_1} = \frac{1}{10}A = \boxed{0.1A}$

$I_2$ also $= \frac{1}{10}A$

b. $I_{tot} = I_1 + I_2 = \frac{1}{5}A$

$P = VI = \frac{1}{5}W = \boxed{0.2W}$

c. current increase - more paths

brightness same; same I and V

2a. $\uparrow N$, $\rightarrow E$ so $\boxed{\text{up}}$

b. still up, but less by sin45.

## HW 8

1a. The wire branches and current is conserved at the junction

b. Because the wires on either side are held at a potential difference of V

c. $I_1 + I_2 = \frac{V}{R_1} + \frac{V}{R_2} = V\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$

d. $I_1 = \frac{120}{12} = 10A$ $\quad I_2 = \frac{120}{8} = 15A$

$I_{tot} = 25A$

e. $P_1 = 120(10) = 1.2kW$ $\quad P_{tot} = 3.0kW$

$P_2 = 120(15) = 1.8kW$

f. More current is supplied; effective total resistance decreases

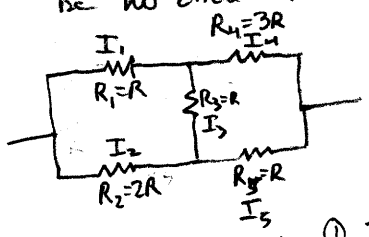
g. $I = \frac{120}{.005} = 24000A$ $\quad I_{tot} = 24015A$

the wire burns up.

h. first, they would have less V across each and not function properly

second, if one was off there would be no circuit or current

2.




Guess $I_3$ going down (will be negative if wrong)

Conservation of current: ① $I_1 = I_3 + I_4$

② $I_2 + I_3 = I_5 \Rightarrow I_2 = I_5 - I_3$

Voltage drop: bottom zig zag top

③ $V = 2RI_2 + RI_5$

④ $V = RI_1 + RI_3 + RI_5$

⑤ $V = RI_1 + 3RI_4$

③ $\frac{V}{R} = 2I_2 + I_5 = 3I_5 - 2I_3 \Rightarrow I_5 = \frac{1}{3}\left(\frac{V}{R} + 2I_3\right)$

⑤ $\frac{V}{R} = I_1 + 3I_4 = I_3 + 4I_4 \Rightarrow I_4 = \frac{1}{4}\left(\frac{V}{R} - I_3\right)$

④ $\frac{V}{R} = I_1 + I_3 + I_5 = 2I_3 + I_4 + I_5 = 2I_3 + \frac{1}{3}\left(\frac{V}{R} + 2I_3\right) + \frac{1}{4}\left(\frac{V}{R} - I_3\right)$

$\frac{V}{R} = \frac{29}{12}I_3 + \frac{7}{12}\frac{V}{R} \Rightarrow \boxed{I_3 = \frac{5}{29}\frac{V}{R} \text{ down}}$

$$\boxed{I_5 = \frac{13}{29}\frac{V}{R} \quad I_4 = \frac{6}{29}\frac{V}{R} \quad I_1 = \frac{11}{29}\frac{V}{R} \quad I_2 = \frac{8}{29}\frac{V}{R} \quad I_{tot} = I_1 + I_2 = \frac{19}{29}\frac{V}{R}}$$

3 a) i) into page ii) out of page iii) down

iv). into page v) down vi) down

b) i) into page ii) into page iii) up and left, 30° W of N iv) no force

c) Rank by angles between v and B:

first iii (90°), then i (60°), then ii (45°) then iv (0°)

4 a) Say 25,000 people, so 25MW plus industry, say $\boxed{35 \text{ MW}}$

b) $P = IV$, $I = \frac{P}{V} = \frac{35 \times 10^6}{500 \times 10^3} = 70A$

c) Use $L = 400$ m

i) $F = BIL\sin 90 = 1.4N$

ii) $\sin\theta = 0$, $F = 0$

iii) $F = 1.4 \sin 45 = 1N$

5. a. $\uparrow B$, $\rightarrow v$ F up. Higher voltage at top, where charge positive

b. $\leftarrow v$ (negatives), $\downarrow B$ F up. Higher V at bottom (negatives go up).